

VANISHING PROPERTIES OF HARMONIC FUNCTIONS IN PLANAR DOMAINS

RUOLIN WANG

1. INTRODUCTION

Let Ω be an open and connected subset of the complex plane $\mathbb{C}$. Consider first an analytic function f in Ω . A well-known result says that if there exists a point $a \in \Omega$ such that

$$(1.1) \quad f^{(n)}(a) = 0 \quad \text{for } n = 0, 1, 2, \dots,$$

then $f(z) = 0$ for all $z \in \Omega$. Furthermore, if a point a in Ω is a limit point for the zero set

$$Z(f) = \{z \in \Omega : f(z) = 0\}$$

of f , then condition (1.1) is satisfied (see for instance Conway [2, Theorem IV.3.7] or Gamelin [3, Section V.7]). This basic result is often referred to as uniqueness of analytic continuation and has a multitude of applications. Conversely, if E is a discrete subset of Ω with no limit point in Ω , then a famous result of Weierstrass ensures existence of an analytic function f in Ω with $Z(f) = E$ (see for instance Conway [2, Theorem VII.5.15]). Together these results describe exactly the zero sets of analytic functions in planar domains.

A purpose of the present paper is to initiate some similar investigations in the context of complex-valued harmonic functions in planar domains. Recall that a function u is said to be harmonic in an open set Ω if it is twice continuously differentiable in Ω (in symbols $u \in C^2(\Omega)$) and satisfies the partial differential equation

$$\Delta u = 0 \quad \text{in } \Omega,$$

where $\Delta = \partial^2/\partial x^2 + \partial^2/\partial y^2$ is the usual Laplacian. Here $z = x + iy$ is the usual decomposition of a complex number $z \in \mathbb{C}$ into its real and imaginary parts $\operatorname{Re}(z) = x \in \mathbb{R}$ and $\operatorname{Im}(z) = y \in \mathbb{R}$, respectively, and the real line $\mathbb{R}$ is considered a subset of the complex plane $\mathbb{C}$ in the usual way. It is a non-trivial course result that harmonic functions are infinitely differentiable: $u \in C^\infty(\Omega)$ if u is harmonic in Ω (see Conway [2, Theorem III.2.30] and [2, Corollary IV.2.12]).

We shall need also the standard complex partial derivatives defined by

$$\partial_z = \frac{1}{2} \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) \quad \text{and} \quad \bar{\partial}_z = \frac{1}{2} \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right),$$

respectively, where $i^2 = -1$ is the imaginary unit and $z = x + iy$ is as above (see for instance Gamelin [3, Section IV.8]). It is an elementary fact that the Laplacian factorizes as $\Delta = 4\partial\bar{\partial}$ using the differential operators ∂ and $\bar{\partial}$.

We point out that the differential operator $\partial + \bar{\partial}$ is the partial derivative with respect to the real component often denoted $\partial/\partial x$. We can think of $\partial + \bar{\partial}$ as a directional derivative or vector field in the direction of the real axis.

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Consider now a complex-valued harmonic function in a planar open set Ω intersecting the real line $\mathbb{R}$. Assume that there exists a point $a \in \Omega \cap \mathbb{R}$ such that

$$(1.2) \quad (\partial + \bar{\partial})^k u(a) = 0 \quad \text{for } k = 0, 1, 2, \dots$$

We prove that then $u(x) = 0$ for every point x in the connected component of $\Omega \cap \mathbb{R}$ containing the point a (see Theorem 2).

The example

$$u(z) = \text{Im}(z), \quad z \in \mathbb{C},$$

shows that condition (1.2) is not demanding enough to ensure vanishing of u outside of the real axis. Furthermore, the set $\Omega \cap \mathbb{R}$ can have several connected components and u can be non-vanishing in components of $\Omega \cap \mathbb{R}$ not containing the point a . An example of this phenomenon is furnished by the argument function $u = \arg(\cdot)$ defined in a slit plane, say,

$$\Omega = \{z \in \mathbb{C} : \text{Re}(z) \neq 0 \text{ or } \text{Im}(z) > 0\},$$

where one has $u(x) = 0$ for $x > 0$ and $u(x) = \pi$ for $x < 0$. For reasons indicated above we consider the analysis of condition (1.2) a main contribution of this paper.

Furthermore, as an direct application of condition (1.2), we first prove a limit point version on the real axis. More precisely, if $a \in \Omega \cap \mathbb{R}$ is a limit point of the set $Z(u) \cap \mathbb{R}$, then condition (1.2) is satisfied and u vanishes on the whole connected component of $\Omega \cap \mathbb{R}$ containing a (see Theorem 3).

We then apply this principle to several classes of planar curves. We show that if the intersection of the zero set $Z(u)$ of a complex-valued harmonic function u with a given curve has a limit point on that curve, then the function vanishes on the whole corresponding curve. For lines, we prove that if $Z(u) \cap L$ has a limit point on a line L , then u vanishes on L (see Theorem 4). We also obtain a stronger result for two non-parallel lines under an additional angle condition (see Theorem 5). For circles C and ellipses E , we show that if $Z(u) \cap C$ or $Z(u) \cap E$ has a limit point on the curve, then u vanishes on the entire curve (see Theorem 6, Theorem 7, and their corresponding corollaries). For parabolas P and hyperbolic arcs H , we prove analogous results: if $Z(u) \cap P$ or $Z(u) \cap H$ has a limit point on the curve, then u vanishes on the whole curve (see Theorem 8 and Theorem 9, together with their corresponding corollaries).

In the entire case, stronger conclusions can be obtained for certain curves. For instance, if $Z(u) \cap C$ or $Z(u) \cap E$ has a limit point on C or E , then the function vanishes identically in $\mathbb{C}$ (see Theorem 6 and Corollary 2). By contrast, vanishing on a parabola or a hyperbola does not in general force identically vanishing in $\mathbb{C}$. However, in the case of hyperbolas we obtain a stronger conclusion: if $Z(u) \cap H$ has a limit point on a hyperbolic arc H , where u is a complex-valued entire harmonic function, then u vanishes on the whole hyperbola $\mathcal{H}$ (see Theorem 10).

This paper is based on joint work with Anders Olofsson. For further information we refer to the paper Khavinson and Shapiro [4] which has also served as a good inspiration. We mention also an earlier bachelor's thesis Wright [9]. We have also benefited from the work of Klintborg and Olofsson [5], where series expansions for generalized harmonic functions are developed, as well as from Olofsson and Wittsten [7], which contains related uniqueness results for weighted harmonic functions. These works provide useful background and context for the results presented in this paper.

2. PRELIMINARIES

We point out some important preliminaries regarding concepts in topology. We denote by

$$B(a, r) = \{z \in \mathbb{C} : |z - a| < r\}$$

the open disc with center $a \in \mathbb{C}$ and radius $r > 0$. An open planar set Ω is a union of discs. A set F is closed in $\mathbb{C}$ if its complement $\mathbb{C} \setminus F$ is open in $\mathbb{C}$. In this way the complex plane $\mathbb{C}$ is a metric space defined by the metric

$$d(z_1, z_2) = |z_1 - z_2|, \quad z_1, z_2 \in \mathbb{C}.$$

A subset E of $\mathbb{C}$ is considered a metric space on its own right by restricting the above metric d to the set $E \times E$. This topologization of E is often referred to as the relative topology.

A metric space X is called connected if it can not be represented as a disjoint union of two non-empty open subsets. A connected component of X is a maximal connected subset of X , that is, a subset X_0 of X is a connected component of X if it is connected and has the property that $X_0 \subset E \subset X$ and E connected implies that $E = X_0$. Since every singleton set is connected, we deduce that connected components are non-empty. A fundamental result in the theory of connectedness says that every topological space can be decomposed as a disjoint union of connected components (see Theorem C in Section 32 in [8]).

Let us discuss connected subsets of $\mathbb{R}$. Recall that a subset E of $\mathbb{C}$ is called convex if for every $z_1, z_2 \in E$ the line segment

$$[z_1, z_2] = \{(1 - t)z_1 + tz_2 : 0 \leq t \leq 1\}$$

is contained in E . A linear combination of the form $(1 - t)z_1 + tz_2$ with $0 \leq t \leq 1$ is commonly called a convex combination of z_1 and z_2 .

Proposition 1. *A subset E of $\mathbb{R}$ is connected if and only if it is convex.*

Proof. We first consider the only if part. Assume E is connected but not convex, then there exists $x, y \in E$, $x < y$ such that there is another point $z \notin E$ with $x < z < y$. Then there is a disconnection of $E = (E \cap (-\infty, z)) \cup (E \cap (z, +\infty))$. Therefore E is not connected, which is a contradiction, hence E is convex.

We next consider the if part. Assume E is a convex set. Then for every $x, y \in E$, $x < y$ we have the interval $[x, y] \subseteq E$. Assume E is not connected. Then there exist U, V disjoint non-empty open sets with $E = U \cup V$. Let points $u \in U$ and $v \in V$, $u < v$ then $[u, v] \subseteq E$, by convexity. Let $z = \sup([u, v] \cap U)$, by openness of U and V , we have $z \notin U$ and $z \notin V$. This contradicts the fact that $z \in E$, hence E is connected. This concludes the proof. $\square$

We point out that the least upper bound property of real numbers is used in an essential way in the above proof.

Let $E \subset \mathbb{R}$. We next define a relation $\sim$ on the set E . We say that $x_1 \in E$ and $x_2 \in E$ are related $x_1 \sim x_2$ if the interval

$$[\min(x_1, x_2), \max(x_1, x_2)]$$

is contained in E . In view of the linear order structure of $\mathbb{R}$ this relation $\sim$ is well-defined. Notice that $x_1 \sim x_2$ means that the interval with endpoints x_1 and x_2 is contained in E .

Proposition 2. *Let $E \subset \mathbb{R}$ and consider the relation $\sim$ defined above. Then the relation $\sim$ is an equivalence relation on E . Furthermore, the partition of E induced by $\sim$ is the decomposition of E into connected components.*

Proof. We first show that the relation $\sim$ is an equivalence relation on E . Clearly, the relation $\sim$ is reflexive and symmetric. We proceed to prove the transitivity property of $\sim$. Assume that $a \sim b$ and $b \sim c$. Then

$$[\min(a, b), \max(a, b)] \subseteq E \quad \text{and} \quad [\min(b, c), \max(b, c)] \subseteq E.$$

Since $\min(b, c) \leq b \leq \max(a, b)$ and $\min(a, b) \leq b \leq \max(b, c)$, we have that

$$[\min(a, b, c), \max(a, b, c)] \subseteq E.$$

As a consequence $[\min(a, c), \max(a, c)] \subseteq E$. Therefore $a \sim c$. This proves transitivity of $\sim$.

The equivalence relation $\sim$ induces a partition of E . We next consider the equivalence class

$$C(x_0) = \{x \in E : x \sim x_0\}$$

of an element $x_0 \in E$.

We first prove that $C(x_0)$ is connected. Let $x_1, x_2 \in C(x_0)$. Without loss of generality we can assume that $x_1 \leq x_2$. Let

$$x = (1 - t)x_1 + tx_2$$

be a convex combination of x_1 and x_2 . We shall prove that $x \in C(x_0)$. Since $x_1, x_2 \in C(x_0)$ we have that $x_1 \sim x_0$ and $x_2 \sim x_0$. By symmetry and transitivity we have that $x_1 \sim x_2$. In view of definition of $\sim$ this means that the interval $[x_1, x_2]$ with end points x_1 and x_2 is contained in E . Since $x \in [x_1, x_2]$ we have that $x \in E$ and $[x_1, x] \subset E$. Therefore $x_1 \sim x$. Since also $x_0 \sim x_1$ we have from transitivity that $x_0 \sim x$. Therefore $x \in C(x_0)$. This proves that every convex combination of x_1 and x_2 belongs to $C(x_0)$. Varying $x_1, x_2 \in C(x_0)$ we have that $C(x_0)$ is convex. By Proposition 1 we have that $C(x_0)$ is connected.

We next prove that $C(x_0)$ is a connected component of E . Assume that $C(x_0) \subset D \subset E$ and D is connected. We prove that $D = C(x_0)$. Let $x \in D$ and observe that D is convex by Proposition 1. Since also $x_0 \in D$ by assumption, we have that the interval with end points x_0 and x is contained in D . Therefore $x_0 \sim x$ and $x \in C(x_0)$. This proves that $D = C(x_0)$.

We next prove that every connected component of E has the form $C(x_0)$ for some $x_0 \in E$. Let D be such a connected component of E . Since D is non-empty there exists $x_0 \in D$. We first prove that $D \subset C(x_0)$. Let $x \in D$. By Proposition 1 the set D is convex so the interval with endpoints x_0 and x is contained in D . Therefore $x_0 \sim x$ and $x \in C(x_0)$. This proves that $D \subset C(x_0)$. Since $C(x_0)$ is connected, we conclude from the maximality property of D that $D = C(x_0)$. $\square$

Notice that the non-empty convex subsets of the set $\mathbb{Q}$ of rational numbers are the singleton subsets. We conclude from Proposition 1 that the connected components of $\mathbb{Q}$ are its singleton subsets. Such a space is sometimes called totally disconnected.

A well-known result says that a function u in a disc $B(a, r)$ is harmonic in $B(a, r)$ if and only if it has a power series expansion of the form

$$(2.1) \quad u(z) = \sum_{k=0}^{\infty} c_k (z - a)^k + \sum_{k=1}^{\infty} c_{-k} (\bar{z} - \bar{a})^k, \quad z \in B(a, r),$$

for some sequence $\{c_k\}_{k=-\infty}^{\infty}$ of complex numbers such that

$$(2.2) \quad \limsup_{|k| \rightarrow \infty} |c_k|^{1/|k|} \leq 1/r.$$

In fact, this follows by a corresponding expansion result for analytic functions combined with existence of a harmonic conjugate. See Conway [2, Theorem IV.2.8] and [2, Theorem III.2.30] for details.

Recall that convergent power series expansions are uniformly convergent on compact subsets of its disc of convergence (see Conway [2, Theorem III.1.3]). Recall also that convergent power series expansions are differentiated termwise (see Conway [2, Proposition III.2.5]). As a consequence of (2.1) we have that $u \in C^\infty(\Omega)$ if u is harmonic in an open set Ω . For similar reason we have that $c_0 = u(a)$,

$$(2.3) \quad c_k = \frac{\partial^k u(a)}{k!} \quad \text{and} \quad c_{-k} = \frac{\bar{\partial}^k u(a)}{k!}$$

for $k \geq 1$, where the c_k 's are as in (2.1).

The following uniqueness result is well-known.

Theorem 1. *Let Ω be an open and connected planar set and $a \in \Omega$. Let u be a complex-valued harmonic function in Ω . Assume that $u(a) = 0$ and*

$$\partial^k u(a) = \bar{\partial}^k u(a) = 0$$

for $k = 1, 2, \dots$. Then $u(z) = 0$ for $z \in \Omega$.

Theorem 1 is proved along similar lines as the corresponding result for analytic functions (see Gamelin [3, Section V.4]). We point out that the series expansion (2.1) can be used as a replacement for the standard power series expansion of an analytic function.

We denote by $\mathbb{Z}$ the set of integers.

Proposition 3. *Let u be a complex-valued harmonic function in $B(a, r)$ of the form (2.1) for some sequence $\{c_k\}_{k=-\infty}^\infty$ of complex numbers satisfying (2.2). Then*

$$(2.4) \quad c_k = \frac{1}{2\pi\rho^{|k|}} \int_{-\pi}^{\pi} u(a + \rho e^{i\theta}) e^{-ik\theta} d\theta$$

for $k \in \mathbb{Z}$, where $0 < \rho < r$.

Proof. From the power series expansion (2.1) we have that

$$u(a + \rho e^{i\theta}) = \sum_{m=0}^{\infty} c_m \rho^m e^{im\theta} + \sum_{m=1}^{\infty} c_{-m} \rho^m e^{-im\theta}.$$

Integrating termwise we have that

$$\begin{aligned} \int_{-\pi}^{\pi} u(a + \rho e^{i\theta}) e^{-ik\theta} d\theta &= \sum_{m=0}^{\infty} c_m \rho^m \int_{-\pi}^{\pi} e^{i(m-k)\theta} d\theta \\ &\quad + \sum_{m=1}^{\infty} c_{-m} \rho^m \int_{-\pi}^{\pi} e^{-i(m+k)\theta} d\theta. \end{aligned}$$

Notice that the change of order of summation and integration is permitted by uniform convergence. From orthogonality of exponential monomials we have that

$$\int_{-\pi}^{\pi} u(a + \rho e^{i\theta}) e^{-ik\theta} d\theta = c_k \rho^{|k|} 2\pi.$$

This completes the proof of the proposition. $\square$

From Proposition 3, it is straightforward to check that the coefficient integral (2.4) can be represented as a line integral by a change of variables $z \rightarrow a + \rho e^{i\theta}$:

$$c_k = \frac{1}{2\pi i} \oint_{|z-a|=\rho} \frac{u(z)}{(z-a)^{k+1}} dz \quad \text{and} \quad c_{-k} = -\frac{1}{2\pi i} \oint_{|z-a|=\rho} \frac{u(z)}{(\bar{z}-a)^{k+1}} d\bar{z},$$

for $k \in \mathbb{N}$ and $0 < \rho < r$, where the integration is traversed in the positive orientation.

3. VANISHING OF HARMONIC FUNCTIONS IN GENERAL DOMAINS

Recall that a function u is said to be harmonic in Ω if $u \in C^2(\Omega)$ and $\partial\bar{\partial}u = 0$ in Ω , where

$$\partial = \frac{1}{2} \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) \quad \text{and} \quad \bar{\partial} = \frac{1}{2} \left(\frac{\partial}{\partial x} - \frac{1}{i} \frac{\partial}{\partial y} \right)$$

are the standard complex partial derivatives.

Lemma 1. *Let u be a complex-valued harmonic function in a planar open set Ω . Let $k \in \mathbb{Z}^+$. Then $(\partial + \bar{\partial})^k u = \partial^k u + \bar{\partial}^k u$ in Ω .*

Proof. By the binomial theorem we know that

$$(\partial + \bar{\partial})^k = \sum_{j=0}^k \binom{k}{j} \partial^j \bar{\partial}^{k-j},$$

where we have used equality of mixed partial derivatives. Since u is harmonic we have that $\partial\bar{\partial}u = 0$ in Ω . Differentiating we have that

$$(\partial + \bar{\partial})^k u = \sum_{j=0}^k \binom{k}{j} \partial^j \bar{\partial}^{k-j} u = \partial^k u + \bar{\partial}^k u$$

in Ω . □

We next consider a harmonic function defined in a neighborhood of a point on the real line.

Lemma 2. *Let u be a complex-valued harmonic function in a disc $B(a, r)$ centered at a point a on the real line $\mathbb{R}$. Then*

$$u(x) = \sum_{k=0}^{\infty} \frac{1}{k!} \partial_1^k u(a) (x - a)^k$$

for $x \in \mathbb{R}$ with $|x - a| < r$, where $\partial_1 = \partial + \bar{\partial}$ is the partial derivative on the real part of the complex variable $z = x + iy$.

Proof. Since u is harmonic in $B(a, r)$ it has a series expansion

$$u(z) = \sum_{k=0}^{\infty} c_k (z - a)^k + \sum_{k=1}^{\infty} c_{-k} (\bar{z} - a)^k$$

convergent for $z \in B(a, r)$. By coefficient formulas (2.3) we have that $c_0 = u(a)$, $c_k = \partial^k u(a)/k!$ and $c_{-k} = \bar{\partial}^k u(a)/k!$ for $k \geq 1$. Restricting to the real axis with $x \in (a - r, a + r)$, we obtain

$$u(x) = c_0 + \sum_{k=1}^{\infty} (c_k + c_{-k})(x - a)^k, \quad x \in (a - r, a + r).$$

In view of Lemma 1 we have that

$$c_k + c_{-k} = \frac{1}{k!} \partial_1^k u(a)$$

for $k = 1, 2, \dots$. In this way we obtain an expansion

$$u(x) = u(a) + \sum_{k=1}^{\infty} \frac{1}{k!} \partial_1^k u(a) (x - a)^k$$

for $x \in (a - r, a + r)$. □

Observe that the differential operator $\partial + \bar{\partial} = \partial/\partial x$ appearing above is the partial derivative on the real part of the complex variable $z = x + iy$.

Theorem 2. *Let u be a complex-valued harmonic function in a planar open set Ω . Let $a \in \Omega \cap \mathbb{R}$. Assume that $u(a) = 0$ and*

$$\partial^k u(a) + \bar{\partial}^k u(a) = 0$$

for $k = 1, 2, \dots$. Then $u(x) = 0$ for every point x in the connected component of $\Omega \cap \mathbb{R}$ containing a .

Proof. For simplicity of notation we set $X = \Omega \cap \mathbb{R}$, which is an open subset of $\mathbb{R}$. Consider the set

$$E = \left\{ x \in X : u(x) = 0 \text{ and } \partial^k u(x) + \bar{\partial}^k u(x) = 0 \text{ for } k = 1, 2, \dots \right\}.$$

Clearly $E \subset X$. We shall prove that E is an open and closed subset of X .

We first prove that E is a closed subset of X . Observe that

$$X \setminus E = \{x \in X : u(x) \neq 0\} \cup \bigcup_{k=1}^{\infty} \{x \in X : \partial^k u(x) + \bar{\partial}^k u(x) \neq 0\}.$$

The set $X \setminus E$ is open in X since it is a union of open subsets of X . This proves that E is a closed subset of X .

We next prove that E is an open subset of X . Let $x_0 \in E$. Since $x_0 \in \Omega$ and Ω is open, there exists $r > 0$ such that $B(x_0, r) \subset \Omega$. By Lemma 2, the restriction of u to the real axis has the expansion

$$u(x) = \sum_{k=0}^{\infty} \frac{1}{k!} \partial_1^k u(x_0) (x - x_0)^k, \quad x \in (x_0 - r, x_0 + r),$$

where $\partial_1 = \partial + \bar{\partial}$. Since $x_0 \in E$, we have that $u(x_0) = 0$. Then Lemma 1 implies that

$$\partial_1^k u(x_0) = \partial^k u(x_0) + \bar{\partial}^k u(x_0) = 0 \quad \text{for all } k \geq 1.$$

Hence, $u(x) = 0$, for $x \in (x_0 - r, x_0 + r)$. Let $k \in \mathbb{Z}^+$. By differentiation we obtain $\partial_1^k u(x) = 0$ for $x \in (x_0 - r, x_0 + r)$, and therefore by Lemma 1,

$$\partial^k u(x) + \bar{\partial}^k u(x) = 0, \quad x \in (x_0 - r, x_0 + r).$$

In this way we conclude that $(x_0 - r, x_0 + r) \subset E$. Since $x_0 \in E$ was arbitrary, it follows that E is open in X .

We now deduce the conclusion of the theorem. Since the set X is open in $\mathbb{R}$, it has the form of a disjoint union of at most countably many disjoint open convex intervals, this is a similar result of [2, Theorem 2.9]. These intervals are the connected components of X . Since the set E is open and closed in X , the set E is a union of connected components of X [2, Theorem 2.7]. Since $a \in E$ by assumption, we conclude that the connected component of X containing the point a is a subset of E . This yields the conclusion of the theorem. $\square$

Remark. We point out that the vanishing condition in Theorem 2 can also be formulated as

$$(\partial + \bar{\partial})^k u(a) = 0 \quad \text{for } k = 0, 1, 2, \dots$$

In fact, this follows from Lemma 1.

In accordance with the notation in the introduction, for a complex-valued harmonic function u on a planar open set Ω we define its zero set by

$$Z(u) = \{z \in \Omega : u(z) = 0\}.$$

A point $z_0 \in \mathbb{C}$ is called a limit point for a planar set E if there exists a sequence $\{z_k\}_{k=1}^{\infty}$ of distinct points from E such that $z_k \rightarrow z_0$ as $k \rightarrow \infty$. Notice that z_0 is a limit point for E if and only if

$$E \cap B(z_0, \delta) \setminus \{z_0\} \neq \emptyset$$

for every $\delta > 0$.

Theorem 3. *Let u be a complex-valued harmonic function in a planar open set Ω . Assume that the set $Z(u) \cap \mathbb{R}$ has a limit point $a \in \Omega \cap \mathbb{R}$. Then $u(x) = 0$ for every x in the connected component of $\Omega \cap \mathbb{R}$ containing a .*

Proof. Let $\partial_1 = \partial + \bar{\partial}$ be the partial derivative on the real part. We shall prove by induction that $\partial_1^m u(a) = 0$ for $m = 0, 1, 2, \dots$. The conclusion of the theorem then follows by Theorem 2. By assumption there exists a sequence $\{x_j\}_{j=1}^\infty$ of distinct points from the set $Z(u) \cap \mathbb{R}$ such that $x_j \rightarrow a$ as $j \rightarrow \infty$. Now $u(x_j) = 0$ for $j = 1, 2, \dots$. Passing to the limit as $j \rightarrow \infty$ we conclude that $u(a) = 0$.

We now turn to the induction step. Assume that $\partial_1^j u(a) = 0$ for $0 \leq j < m$, where $m \geq 1$. We proceed to prove that $\partial_1^m u(a) = 0$. Since $a \in \Omega$ and Ω is an open planar set, there exists $r > 0$ such that $B(a, r) \subset \Omega$. Consider the power series expansion of u from Lemma 2. From the induction hypothesis we deduce that

$$(3.1) \quad u(x) = \sum_{k=m}^{\infty} \frac{1}{k!} \partial_1^k u(a) (x-a)^k$$

for $x \in (a-r, a+r)$.

We now prove that $\partial_1^m u(a) = 0$. Let

$$g(x) = \sum_{k=0}^{\infty} \frac{1}{(k+m)!} \partial_1^{k+m} u(a) (x-a)^k$$

for $x \in (a-r, a+r)$. From (3.1) we have that

$$u(x) = (x-a)^m g(x)$$

for $x \in (a-r, a+r)$. Evaluating at the point x_j we have that

$$0 = u(x_j) = (x_j - a)^m g(x_j)$$

for $j = 1, 2, \dots$. Since $x_j \neq a$, we conclude that $g(x_j) = 0$ for $j = 1, 2, \dots$. Passing to the limit we obtain that $g(a) = 0$. This proves that $\partial_1^m u(a) = 0$. The claim that $\partial_1^m u(a) = 0$ for $m = 0, 1, 2, \dots$ now follows by the principle of induction. $\square$

4. VANISHING OF ENTIRE HARMONIC FUNCTIONS

By an automorphism of the complex plane we understand a function $\tau = \tau_{a,b}$ of the form

$$\tau_{a,b}(z) = az + b, \quad z \in \mathbb{C},$$

for some $a, b \in \mathbb{C}$ with $a \neq 0$. The set of all such automorphisms τ is denoted $\text{Aut}(\mathbb{C})$. If $|a| = 1$, then $\tau_{a,b}$ is called a rigid motion of the plane. The set $\text{Aut}(\mathbb{C})$ is a group. The group operation is composition of functions. A well-known result from analysis says that every analytic bijection of the complex plane has the form $\tau_{a,b}$ for some a, b as above (see for instance Lang [6, Theorem V.3.3]). The proof follows from the famous Casorati–Weierstrass theorem.

By a line in the complex plane we understand a set L of the form

$$L = \tau(\mathbb{R}) = \left\{ z \in \mathbb{C} : z = \tau(x) \text{ for some } x \in \mathbb{R} \right\}$$

for some $\tau \in \text{Aut}(\mathbb{C})$.

Theorem 4. *Let u be a complex-valued entire harmonic function. Let L be a line in $\mathbb{C}$ and assume that the set $Z(u) \cap L$ has a limit point. Then $u(z) = 0$ for all $z \in L$.*

Proof. We put the line L of the form $L = \tau(\mathbb{R})$ for some $\tau \in \text{Aut}(\mathbb{C})$. Consider the function $v = u \circ \tau$. Now v is an entire harmonic function. Notice that $\tau(Z(v)) = Z(u)$. By assumption the set $Z(v) \cap \mathbb{R}$ has a limit point. From Theorem 3 we conclude that $v(x) = 0$ for $x \in \mathbb{R}$. This gives the conclusion that $u(z) = 0$ for all $z \in L$. $\square$

The harmonic polynomial

$$v(z) = \text{Im}(z), \quad z \in \mathbb{C},$$

vanishes precisely on the real line $\mathbb{R}$. Composing with an automorphism τ from $\text{Aut}(\mathbb{C})$ we can produce a harmonic polynomial $u = v \circ \tau^{-1}$ vanishing precisely on the line $L = \tau(\mathbb{R})$.

Let L_1 and L_2 be lines in $\mathbb{C}$. Suppose

$$L_j = \tau_j(\mathbb{R}), \quad j = 1, 2,$$

for automorphisms $\tau_j(x) = a_j x + b_j$ with $a_j \neq 0$. Let $\theta_j \in [0, \pi)$ be such that

$$\theta_j \equiv \arg(a_j) \pmod{\pi},$$

that is, θ_j represents the direction of the line L_j .

We define the angle between L_1 and L_2 by

$$\angle(L_1, L_2) = \min\{|\theta_1 - \theta_2|, \pi - |\theta_1 - \theta_2|\} \in [0, \pi/2].$$

We say that L_1 and L_2 are *parallel* if $\theta_1 = \theta_2$, or equivalently if $\angle(L_1, L_2) = 0$. Otherwise, the lines are said to be *non-parallel*.

Theorem 5. *Let u be a complex-valued entire harmonic function. Let L_1 and L_2 be two non-parallel lines in $\mathbb{C}$. Assume that for $j = 1, 2$, the set $Z(u) \cap L_j$ has a limit point $a_j \in L_j$. If the angle $\angle(L_1, L_2)$ is not a rational multiple of π , then $u(z) = 0$ for all $z \in \mathbb{C}$.*

Proof. By assumption, for $j = 1, 2$ the set $Z(u) \cap L_j$ has a limit point. Hence by Theorem 4, $u(z) = 0$ for all $z \in L_1 \cup L_2$. Since L_1 and L_2 are non-parallel, their intersection consists of a single point $\{a\} = L_1 \cap L_2$. Let $\tau_j \in \text{Aut}(\mathbb{C})$ be such that $L_j = \tau_j(a_j x + b_j)$, for $j = 1, 2$ with $a_j \neq 0$. Since $a \in L_j$, we have

$$L_j = \{a + a_j x : x \in \mathbb{R}\}, \quad j = 1, 2.$$

Since u vanishes on L_j , we have $u(a + a_j x) = 0$ for all $x \in \mathbb{R}$. Since $u(a + a_j x) = 0$ for all $x \in \mathbb{R}$, we have

$$\frac{d^m}{dx^m} u(a + a_j x) = 0, \quad x \in \mathbb{R}, \quad m \geq 1.$$

A differentiation by the chain rule yields

$$\frac{d^m}{dx^m} u(a + a_j x) = (a_j \partial + \bar{a}_j \bar{\partial})^m u(a + a_j x).$$

Evaluating at $x = 0$ and by Lemma 1, we have

$$(a_j \partial + \bar{a}_j \bar{\partial})^m u(a) = a_j^m \partial^m u(a) + \bar{a}_j^m \bar{\partial}^m u(a), \quad j = 1, 2.$$

Thus for each $m \geq 1$ we obtain the homogeneous linear system

$$\begin{aligned} a_1^m \partial^m u(a) + \bar{a}_1^m \bar{\partial}^m u(a) &= 0, \\ a_2^m \partial^m u(a) + \bar{a}_2^m \bar{\partial}^m u(a) &= 0. \end{aligned}$$

Assume that the determinant of the system above is zero, then $a_1^m \bar{a}_2^m = a_2^m \bar{a}_1^m$. This implies

$$\left(\frac{a_2}{a_1}\right)^m = \overline{\left(\frac{a_2}{a_1}\right)^m}.$$

Hence $(a_2/a_1)^m$ is real, so

$$m(\theta_2 - \theta_1) \in \pi\mathbb{Z},$$

where $\theta_j \equiv \arg(a_j) \pmod{\pi}$. It follows that $\angle(L_1, L_2)$ is a rational multiple of π , which contradicts the assumption. Therefore, the determinant is nonzero, and we conclude that $\partial^m u(a) = 0$ and $\bar{\partial}^m u(a) = 0$ for all $m \geq 1$. Since also $u(a) = 0$, Theorem 1 yields $u(z) = 0$ for all $z \in \mathbb{C}$. This completes the proof. $\square$

Remark. The conclusion of Theorem 5 can also be obtained by using the local series expansion of a harmonic function centered at a with $z = a + te^{i\theta_j}$, $j = 1, 2$ which gives identities for the coefficients c_k and c_{-k} . The assumption that $\angle(L_1, L_2)$ is not a rational multiple of π forces all coefficients to vanish. Hence $u(z) = 0$ in a neighborhood of a , and by Theorem 1 we conclude that $u(z) = 0$ for all $z \in \mathbb{C}$.

We point out that the non-parallel assumption in Theorem 5 is essential in the example below.

Example 1. Consider the real-valued entire harmonic function

$$u(z) = \operatorname{Re}(e^z), \quad z \in \mathbb{C}.$$

Writing $z = x + iy$ we have

$$u(z) = \operatorname{Re}(e^{x+iy}) = \operatorname{Re}(e^x(\cos y + i \sin y)) = e^x \cos y.$$

Hence $u(z) = 0$ if and only if $\cos y = 0$, that is u vanishes on each of the parallel lines

$$L_k = \left\{ x + i\left(\frac{\pi}{2} + k\pi\right) : x \in \mathbb{R} \right\}, \quad k \in \mathbb{Z}.$$

Clearly $u(0) = 1$, therefore u is not identically zero. This shows that if one allows parallel lines, then the conclusion $u(z) = 0$ for all $z \in \mathbb{C}$ need not hold.

We next point out that vanishing on several intersecting lines does not in general force an entire harmonic function to vanish identically.

Example 2. Fix an integer $m \geq 1$ and consider the real-valued entire harmonic function

$$u(z) = \operatorname{Im}(z^m), \quad z \in \mathbb{C}.$$

Writing $z = re^{i\theta}$ we obtain

$$u(z) = \operatorname{Im}(r^m e^{im\theta}) = r^m \sin(m\theta).$$

Hence $u(z) = 0$ if and only if $r = 0$ or $\theta = \frac{k\pi}{m}$ for some $k \in \mathbb{Z}$. Therefore u vanishes on each of the m distinct lines through the origin

$$L_k = \{ te^{ik\pi/m} : t \in \mathbb{R} \}, \quad k = 0, 1, \dots, m-1.$$

For $k \neq \ell$, $\angle(L_k, L_\ell)$ is a rational multiple of π . Thus Theorem 5 does not apply to this configuration of lines. u is clearly not identically zero. For instance, let $m = 2$

$$u(1+i) = \operatorname{Im}((1+i)^2) \neq 0.$$

We first recall a version of the maximum principle (see for instance [6, Theorem VIII.3.4(b)]).

Lemma 3. *Let Ω be a bounded and connected planar open set, and let u be a complex-valued harmonic function in a neighbourhood of $\bar{\Omega}$. Assume that $u(z) = 0$ for all $z \in \partial\Omega$. Then $u(z) = 0$ for all $z \in \Omega$. Moreover, if u is entire harmonic, then $u(z) = 0$ for all $z \in \mathbb{C}$.*

Proof. Let $u = v + iw$, where $v = \operatorname{Re}(u)$ and $w = \operatorname{Im}(u)$ are real-valued harmonic functions. Since $u = 0$ on $\partial\Omega$, we have both $v = 0$ and $w = 0$ on $\partial\Omega$. By the famous maximum principle for real-valued harmonic functions, both v and w vanish identically in Ω . Hence $u = 0$ in Ω .

If u is entire harmonic, then u vanishes on the non-empty open set Ω . Fix $a \in \Omega$, since $u = 0$ in a neighbourhood of a , we have all coefficients in the expansion of u of the form in (2.1) vanish, that is

$$u(a) = 0, \quad \partial^k u(a) = 0, \quad \bar{\partial}^k u(a) = 0 \quad \text{for all } k \geq 1.$$

Therefore, by Theorem 1, we conclude that $u(z) = 0$ for all $z \in \mathbb{C}$. $\square$

We denote by $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ the unit circle in the complex plane.

Theorem 6. *Let u be a complex-valued entire harmonic function. If the set $Z(u) \cap \mathbb{T}$ has a limit point, then $u(z) = 0$ for all $z \in \mathbb{C}$.*

Proof. We consider the function

$$v(z) = u(e^{iz}), \quad z \in \mathbb{C}.$$

The function v is entire harmonic. By assumption the set $Z(v) \cap \mathbb{R}$ has a limit point. By Theorem 3 we obtain that $v(x) = 0$ for all $x \in \mathbb{R}$. Returning to the function u , this means that $u(e^{ix}) = 0$ for all $x \in \mathbb{R}$. Therefore $u(z) = 0$ for all $z \in \mathbb{T}$. The conclusion then follows by Lemma 3. $\square$

Remark. The conclusion of Theorem 6 can also be obtained by using the coefficient formula in Proposition 3. Since u vanishes on $\mathbb{T}$, we obtain $c_k = 0$ for all $k \in \mathbb{Z}$. Hence all coefficients in the expansion of u of the form (2.1) vanish, and therefore $u(z) = 0$ for all $z \in \mathbb{D}$ with

$$u(0) = 0, \quad \partial^m u(0) = 0, \quad \bar{\partial}^m u(0) = 0 \quad \text{for all } m \geq 1.$$

By Theorem 1 we conclude that $u(z) = 0$ for all $z \in \mathbb{C}$.

We next generalize Theorem 6 to allow for a general circle. A circle in the complex plane is a set C of the form $C = \tau(\mathbb{T})$ for some $\tau \in \operatorname{Aut}(\mathbb{C})$, where $\mathbb{T}$ is the unit circle in the complex plane.

Corollary 1. *Let u be a complex-valued entire harmonic function and let C be a circle in the complex plane. If the set $Z(u) \cap C$ has a limit point, then $u(z) = 0$ for all $z \in \mathbb{C}$.*

Proof. Write the circle C in the form $C = \tau(\mathbb{T})$ for some $\tau \in \operatorname{Aut}(\mathbb{C})$. Consider the composition of functions $v = u \circ \tau$. Then v is an entire harmonic function in $\mathbb{C}$. Hence

$$Z(v) \cap \mathbb{T} = \tau^{-1}(Z(u) \cap C).$$

Since $Z(u) \cap C$ has a limit point and τ^{-1} is continuous, it follows that $Z(v) \cap \mathbb{T}$ has a limit point. By Theorem 6 we conclude that $v(z) = 0$ for all $z \in \mathbb{C}$. Returning to the function u , this means that $u(\tau(z)) = 0$ for all $z \in \mathbb{C}$. Since τ is onto, we conclude that $u(z) = 0$ for all $z \in \mathbb{C}$. $\square$

We consider a normalized ellipse of the form

$$(4.1) \quad E_r = \left\{ x + iy \in \mathbb{C} : x, y \in \mathbb{R} \text{ and } \left(\frac{x}{r} \right)^2 + (ry)^2 = 1 \right\}$$

for some $r > 0$.

We next discuss vanishing conditions on ellipses.

Theorem 7. *Let u be a complex-valued entire harmonic function. Let E_r be a normalized ellipse of the form (4.1) for some $r > 0$. If the set $Z(u) \cap E_r$ has a limit point, then $u(z) = 0$ for all $z \in \mathbb{C}$.*

Proof. Introduce the auxiliary function $\phi(z) = r \cos(z) + i \frac{1}{r} \sin(z)$ for $z \in \mathbb{C}$, where

$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2} \quad \text{and} \quad \sin(z) = \frac{e^{iz} - e^{-iz}}{2i}$$

are the entire analytic cosine and sine functions. We consider the composition $v = u \circ \phi$, that is,

$$v(z) = u(r \cos(z) + i \frac{1}{r} \sin(z)), \quad z \in \mathbb{C}.$$

Then v is an entire harmonic function in $\mathbb{C}$. Notice that the set $E_r = \phi(\mathbb{R})$. By assumption, the set $Z(v) \cap \mathbb{R}$ has a limit point. By Theorem 3, it follows that $v(x) = 0$, for all $x \in \mathbb{R}$. Returning to the function u , this means that

$$u(r \cos x + i \frac{1}{r} \sin x) = 0 \quad \text{for all } x \in \mathbb{R}.$$

Therefore $u(z) = 0$ for all $z \in E_r$. Let Ω_r be the ellipse disc enclosed by the ellipse E_r . Since u vanishes on $\partial\Omega_r = E_r$ and u is entire harmonic, then by Lemma 3 implies that $u(z) = 0$ for all $z \in \mathbb{C}$. $\square$

We next generalize Theorem 7 to allow for a general ellipse. By an ellipse in the complex plane we understand a set E of the form $E = \tau(E_r)$ for some $\tau \in \text{Aut}(\mathbb{C})$ and $r > 0$, where E_r is as in (4.1).

Corollary 2. *Let u be a complex-valued entire harmonic function. Let E be an ellipse in the complex plane. If the set $Z(u) \cap E$ has a limit point, then $u(z) = 0$ for all $z \in \mathbb{C}$.*

Proof. Write $E = \tau(E_r)$ for some $\tau \in \text{Aut}(\mathbb{C})$ and $r > 0$. Define

$$v(z) = u(\tau(z)), \quad z \in \mathbb{C}.$$

Since u is entire harmonic and τ is an analytic automorphism of $\mathbb{C}$, the function v is entire harmonic. By assumption

$$Z(v) \cap E_r = \tau^{-1}(Z(u) \cap E).$$

Since $Z(u) \cap E$ has a limit point and τ^{-1} is continuous, it follows that $Z(v) \cap E_r$ has a limit point. By Theorem 7, we conclude that $v(z) = 0$ for all $z \in \mathbb{C}$. Returning to u , this means that $u(\tau(z)) = 0$ for all $z \in \mathbb{C}$. Since τ is onto, we conclude that $u(z) = 0$ for all $z \in \mathbb{C}$. $\square$

We next distinguish the difference between local and global vanishing by the following example.

Example 3. Consider the real-valued entire harmonic function

$$v(z) = \text{Im}z, \quad z \in \mathbb{C},$$

and let

$$\tau(z) = r \cos z + i \frac{1}{r} \sin z, \quad z \in \mathbb{C},$$

for some $r > 0$. Then $\tau(\mathbb{R}) = E_r$. Since $\tau'(0) \neq 0$ and $\tau(0) = r$, by the inverse function theorem (See for instance [6, Theorem II.6.1(c)]) there exist neighborhoods U of 0 and V of r such that $\tau : U \rightarrow V$ is biholomorphic. Write

$$u(w) = v(\tau^{-1}(w)) = \text{Im}(\tau^{-1}(w)), \quad w \in V.$$

Then u is harmonic in V , and for $w \in V \cap E_r$ we have $u(w) = 0$. Thus u is a nontrivial harmonic function, locally defined near the ellipse, which vanishes on a nonempty elliptic subarc. Notice that u is not identically zero, since U is an open neighborhood of 0, there exists $z_0 \in U$ with $\text{Im}z_0 \neq 0$. Let $w_0 = \tau(z_0) \in V$. Then

$$u(w_0) = \text{Im}(\tau^{-1}(w_0)) = \text{Im}z_0 \neq 0.$$

We denote the standard parabola in the complex plane by the set

$$(4.2) \quad P_k = \{x + iy \in \mathbb{C} : x, y \in \mathbb{R} \text{ and } y = kx^2\}$$

for some $k \in \mathbb{R}$.

We next discuss vanishing conditions on parabolas.

Theorem 8. *Let P_k be a standard parabola of the form (4.2) for some $k \in \mathbb{R}$. Let u be a complex-valued harmonic function in an open set Ω containing P_k . If the set $Z(u) \cap P_k$ has a limit point, then $u(z) = 0$ for all $z \in P_k$.*

Proof. Consider the polynomial

$$\phi(z) = z + ikz^2, \quad z \in \mathbb{C},$$

and let

$$v(z) = u(\phi(z)), \quad z \in \mathbb{C}.$$

Since u is harmonic in Ω and ϕ is entire analytic, the function v is harmonic in $\phi^{-1}(\Omega)$. Notice that $\phi(\mathbb{R}) = P_k$. By assumption the set $Z(v) \cap \mathbb{R}$ has a limit point. Hence, by Theorem 3, we obtain that $v(x) = 0$ for all $x \in \mathbb{R}$. Returning to the function u , this means that

$$u(x + ikx^2) = 0 \quad \text{for all } x \in \mathbb{R}.$$

Therefore $u(z) = 0$ for all $z \in P_k$. $\square$

We next generalize Theorem 8 to allow for a general parabola. By a parabola in the complex plane we understand a set P of the form $P = \tau(P_k)$ for some $\tau \in \text{Aut}(\mathbb{C})$ and $k \in \mathbb{R}$, where P_k is as in (4.2).

Corollary 3. *Let P be a parabola in the complex plane. Let u be a complex-valued harmonic function in an open set Ω containing P . If the set $Z(u) \cap P$ has a limit point, then $u(z) = 0$ for all $z \in P$.*

Proof. Write $P = \tau(P_k)$ for some $\tau \in \text{Aut}(\mathbb{C})$ and $k \in \mathbb{R}$. Let

$$v(z) = u(\tau(z)), \quad z \in \tau^{-1}(\Omega).$$

Since u is harmonic in Ω and τ is an analytic automorphism of $\mathbb{C}$, the function v is harmonic in $\tau^{-1}(\Omega)$. Notice that $P_k \subset \tau^{-1}(\Omega)$. By assumption,

$$Z(v) \cap P_k = \tau^{-1}(Z(u) \cap P).$$

Since $Z(u) \cap P$ has a limit point and τ^{-1} is continuous, it follows that $Z(v) \cap P_k$ has a limit point. By Theorem 8, we conclude that $v(z) = 0$ for all $z \in P_k$. Returning to u , this means that

$$u(\tau(z)) = 0 \quad \text{for all } z \in P_k.$$

Since $\tau(P_k) = P$, we conclude that $u(z) = 0$ for all $z \in P$. $\square$

Remark. For entire harmonic functions vanishing on parabola we cannot deduce that it vanishes completely on $\mathbb{C}$. Substituting $z = x + ikx^2$ and $\bar{z} = x - ikx^2$ into the power series expansion as in (2.1) centering at the origin, where the radius of convergence is infinite. We then have

$$0 = u(x + ikx^2) = c_0 + \sum_{n \geq 1} c_n (x + ikx^2)^n + \sum_{n \geq 1} c_{-n} (x - ikx^2)^n.$$

Expanding and comparing coefficients of powers of x yields

$$\begin{aligned} c_0 &= 0, \\ c_1 + c_{-1} &= 0, \\ c_2 + c_{-2} + ik(c_1 - c_{-1}) &= 0, \\ c_3 + c_{-3} + 2ik(c_2 - c_{-2}) &= 0, \\ c_4 + c_{-4} + 3ik(c_3 - c_{-3}) - k^2(c_2 + c_{-2}) &= 0, \end{aligned}$$

with similar identities at higher orders. The system is underdetermined and does not force all coefficients to vanish. We consider this discussion as heuristic since we did not analyze the growth of the c_k 's.

Recall that a function $u : \mathbb{C} \rightarrow \mathbb{C}$ is said to be globally Lipschitz if there exists a constant $L > 0$ such that

$$|u(z) - u(w)| \leq L|z - w|, \quad \text{for all } z, w \in \mathbb{C}.$$

We next show that in the entire setting, vanishing on a parabola with a suitable growth condition forces the function to vanish identically on $\mathbb{C}$.

Proposition 4. *Let P be a parabola in the complex plane. Let u be a complex-valued entire harmonic function on $\mathbb{C}$. Assume that u is globally Lipschitz on $\mathbb{C}$ and that the set $Z(u) \cap P$ has a limit point on P . Then $u(z) = 0$ for all $z \in \mathbb{C}$.*

Proof. Since $Z(u) \cap P$ has a limit point, therefore by Corollary 3 we have that $u(z) = 0$ for all $z \in P$. Since u is globally Lipschitz on $\mathbb{C}$, its first-order partial derivatives are bounded on $\mathbb{C}$. As u is entire harmonic, the functions $\partial u / \partial x$ and $\partial u / \partial y$ are also entire harmonic. By Liouville's theorem for harmonic functions [1, Chap. 4, Sec. 2.3], both $\partial u / \partial x$ and $\partial u / \partial y$ are constant. Hence by integration, u is an affine function of the form

$$u(x + iy) = ax + by + c$$

for some constants $a, b, c \in \mathbb{C}$.

Write $P = \tau(P_k)$ for some $\tau \in \text{Aut}(\mathbb{C})$ and $k \in \mathbb{R}$, where P_k is as in (4.2) for some $k \neq 0$. Consider the composition

$$v(z) = u(\tau(z)), \quad z \in \mathbb{C}.$$

Then v is also an affine function, say

$$v(x + iy) = a'x + b'y + c'$$

for some constants $a', b', c' \in \mathbb{C}$. Since u vanishes on P , it follows that v vanishes on P_k . Thus

$$0 = v(x + ikx^2) = a'x + b'kx^2 + c', \quad \text{for all } x \in \mathbb{R}.$$

Hence the polynomial $b'kx^2 + a'x + c'$ must vanish identically for all $x \in \mathbb{R}$. Since $k \neq 0$, therefore we have

$$a' = 0, \quad b' = 0, \quad c' = 0.$$

Hence $v(z) = 0$ for all $z \in \mathbb{C}$, and therefore $u(z) = 0$ for all $z \in \mathbb{C}$. $\square$

We next illustrate that in the non-entire setting vanishing on a parabola does not in general force a harmonic function to vanish identically.

Example 4. Fix $c > 0$ and define the domain

$$\Omega = \mathbb{C} \setminus (-\infty, 0],$$

that is, the complex plane slit along the negative real axis. Let $\sqrt{\cdot}$ denote the principal branch of the square root on Ω , so that $\sqrt{\Omega} = \mathbb{C}_+$ is the right half-plane. Consider the real-valued function

$$u(z) = \operatorname{Re}(\sqrt{z}) - c, \quad z \in \Omega.$$

Now the set $P_c = \{(c+it)^2 : t \in \mathbb{R}\}$ is a parabola in the complex plane and $P_c \subset \Omega$. Notice that if $z = (c+it)^2 \in P_c$, then $\sqrt{z} = c+it$ for $z \in \Omega$. Therefore

$$u(z) = \operatorname{Re}(c+it) - c = 0.$$

Thus $P_c \subset Z(u)$. Conversely, if $u(z) = 0$, then $\operatorname{Re}(\sqrt{z}) = c$, so $\sqrt{z} = c+it$ for some $t \in \mathbb{R}$. Hence

$$z = (c+it)^2 \in P_c.$$

Therefore $Z(u) \subset P_c$, and we have $Z(u) = P_c$.

We define the normalized hyperbola in the complex plane by

$$(4.3) \quad \mathcal{H}_r = \left\{ x+iy \in \mathbb{C} : x, y \in \mathbb{R} \text{ and } \left(\frac{x}{r}\right)^2 - (ry)^2 = 1 \right\}$$

for some $r > 0$. The set $\mathcal{H}_r$ has two connected components, corresponding to the two branches of the hyperbola. We denote them by

$$(4.4) \quad H_r^+ = \{z \in \mathcal{H}_r : \operatorname{Re}(z) > 0\}$$

and

$$H_r^- = \{z \in \mathcal{H}_r : \operatorname{Re}(z) < 0\}.$$

Each of the sets H_r^+ and H_r^- will be referred to as a normalized hyperbolic arc.

Theorem 9. *Let H_r^+ be a normalized hyperbola arc of form (4.4) for some $r > 0$. Let u be a complex-valued harmonic function in an open set Ω containing H_r^+ . Assume that the set $Z(u) \cap H_r^+$ has a limit point. Then $u(z) = 0$ for all $z \in H_r^+$.*

Proof. Introduce the auxiliary function $\phi(z) = r \cosh z + i \frac{1}{r} \sinh z$ for $z \in \mathbb{C}$, where

$$\cosh(z) = \frac{e^z + e^{-z}}{2} \quad \text{and} \quad \sinh(z) = \frac{e^z - e^{-z}}{2}$$

are the entire analytic hyperbolic cosine and sine functions. We consider the composition $v = u \circ \phi$, that is,

$$v(z) = u(r \cosh z + i \frac{1}{r} \sinh z), \quad z \in \mathbb{C}.$$

Then v is harmonic in $\phi^{-1}(\Omega)$. Consider

$$H_r^+ = \{r \cosh(x) + i \frac{1}{r} \sinh(x) : x \in \mathbb{R}\}.$$

Notice that $H_r^+ = \phi(\mathbb{R}) \subset \Omega$. By assumption, the set $Z(v) \cap \mathbb{R}$ has a limit point. By Theorem 3, it follows that $v(x) = 0$ for all $x \in \mathbb{R}$. Returning to the function u , this means that

$$u(r \cosh x + i \frac{1}{r} \sinh x) = 0, \quad \text{for all } x \in \mathbb{R}.$$

Therefore $u(z) = 0$ for all $z \in H_r^+$. $\square$

We next generalize Theorem 9 to allow for a general hyperbolic arc. By a hyperbolic arc in the complex plane we understand a set H of the form $H = \tau(H_r^+)$ for some $\tau \in \operatorname{Aut}(\mathbb{C})$ and $r > 0$, where H_r^+ is as in (4.4).

Corollary 4. *Let H be a parabolic arc in the complex plane. Let u be a complex-valued harmonic function in an open set Ω containing H . If the set $Z(u) \cap H$ has a limit point, then $u(z) = 0$ for all $z \in H$.*

Proof. Write $H = \tau(H_r^+)$ for some $\tau \in \text{Aut}(\mathbb{C})$ and $r > 0$. Let

$$v(z) = u(\tau(z)), \quad z \in \tau^{-1}(\Omega).$$

Since u is harmonic in Ω and τ is an analytic automorphism of $\mathbb{C}$, the function v is harmonic in $\tau^{-1}(\Omega)$. By assumption,

$$Z(v) \cap H_r^+ = \tau^{-1}(Z(u) \cap H).$$

Since $Z(u) \cap H$ has a limit point and τ^{-1} is continuous, it follows that $Z(v) \cap H_r^+$ has a limit point. By Theorem 9, we conclude that $v(z) = 0$ for all $z \in H_r^+$. Returning to u , this means that

$$u(\tau(z)) = 0 \quad \text{for all } z \in H_r^+.$$

Since $\tau(H_r^+) = H$, we conclude that $u(z) = 0$ for all $z \in H$. $\square$

We next give a corresponding example for hyperbolic arcs, showing that local vanishing on a hyperbola does not imply global vanishing.

Example 5. Fix $c \in (0, \pi/2)$ and define the domain

$$\Omega = \mathbb{C} \setminus (i(-\infty, -1] \cup i[1, \infty)),$$

that is, the complex plane slit along the imaginary axis from i to $i\infty$ and from $-i$ to $-i\infty$. Consider the principal branch of the inverse hyperbolic sine

$$\sinh^{-1} z = \log(z + \sqrt{z^2 + 1})$$

on Ω . It is well-known that (See for instance [1, Chap. 3, Sec. 2.3]) the principal branch of $\sinh^{-1}$ maps Ω conformally onto the strip

$$\{w \in \mathbb{C} : -\pi/2 < \text{Im} w < \pi/2\}.$$

Then we consider the real-valued function

$$u(z) = \text{Im}(\sinh^{-1} z) - c, \quad z \in \Omega.$$

Now the set $H_c^+ = \{\sinh(t + ic) : t \in \mathbb{R}\}$ is a hyperbolic arc in the complex plane and $H_c^+ \subset \Omega$. If $z = \sinh(t + ic)$ with $t \in \mathbb{R}$, then $\sinh^{-1} z = t + ic$, and hence

$$u(z) = \text{Im}(t + ic) - c = 0.$$

Thus $H_c^+ \subset Z(u)$. Conversely, if $u(z) = 0$, then $\text{Im}(\sinh^{-1} z) = c$, so $\sinh^{-1} z = t + ic$ for some $t \in \mathbb{R}$. Hence $z \in H_c^+$. Therefore $Z(u) \subset H_c^+$ and we have $Z(u) = H_c^+$.

We next give an example showing that local vanishing on one branch of the hyperbola does not force vanishing on the other branch.

Example 6. Consider the real-valued entire harmonic function

$$v(z) = \text{Im} z, \quad z \in \mathbb{C},$$

and let

$$\phi(z) = r \cosh z + i \frac{1}{r} \sinh z, \quad z \in \mathbb{C},$$

for some $r > 0$. Then

$$\phi(\mathbb{R}) = H_r^+ \quad \text{and} \quad \phi(x + i\pi) = -r \cosh x - i \frac{1}{r} \sinh x \in H_r^-, \quad x \in \mathbb{R}.$$

Since

$$\phi'(0) = \frac{i}{r} \neq 0 \quad \text{and} \quad \phi'(i\pi) = -\frac{i}{r} \neq 0,$$

the inverse function theorem (See for instance [6, Theorem II.6.1(c)]) yields neighborhoods U_0 of 0 and U_π of $i\pi$, together with neighborhoods V_0 of r and V_π of $-r$, such that

$$\phi : U_0 \rightarrow V_0 \quad \text{and} \quad \phi : U_\pi \rightarrow V_\pi$$

are biholomorphic. We may assume that $V_0 \cap V_\pi = \emptyset$ by shrinking if necessary. Let

$$u(w) = \begin{cases} \operatorname{Im}((\phi|_{U_0})^{-1}(w)), & w \in V_0, \\ \operatorname{Im}((\phi|_{U_\pi})^{-1}(w)), & w \in V_\pi. \end{cases}$$

Then u is harmonic on the open set $V_0 \cup V_\pi$. If $w \in V_0 \cap H_r^+$, then $w = \phi(x)$ for some $x \in \mathbb{R}$, and hence $u(w) = \operatorname{Im}(x) = 0$. Thus u vanishes on a local arc of the right branch. On the other hand, if $w \in V_\pi \cap H_r^-$, then $w = \phi(x + i\pi)$ for some $x \in \mathbb{R}$, and therefore

$$u(w) = \operatorname{Im}(x + i\pi) = \pi \neq 0.$$

Thus u is a nontrivial harmonic function, locally defined near the hyperbola, which vanishes on a local arc of one branch but not on the other branch.

We next turn to the entire case. Corollary 4 discusses vanishing on one hyperbolic arc. For entire harmonic functions one obtains a stronger conclusion. By a general hyperbola in the complex plane we understand a set $\mathcal{H}$ of the form $\mathcal{H} = \tau(\mathcal{H}_r)$ for some $\tau \in \operatorname{Aut}(\mathbb{C})$ and $r > 0$, where $\mathcal{H}_r$ is as in (4.3).

Theorem 10. *Let u be a complex-valued entire harmonic function, and let $\mathcal{H}$ be a hyperbola in the complex plane. Assume that for one hyperbolic arc H of $\mathcal{H}$, the set $Z(u) \cap H$ has a limit point. Then $u(z) = 0$ for all $z \in \mathcal{H}$.*

Proof. Since H is a hyperbolic arc, by Corollary 4 we have that $u(z) = 0$ for all $z \in H$. By definition of a hyperbola in the complex plane, we can write $\mathcal{H} = \tau(\mathcal{H}_r)$, where $\mathcal{H}_r$ is as in (4.3). Replacing u by $u \circ \tau$, we may reduce to the normalized case $\mathcal{H} = \mathcal{H}_r$. Replacing x by $-x$ if necessary, we may also assume that

$$H = H_r^+ = \{r \cosh x + i \frac{1}{r} \sinh x : x \in \mathbb{R}\}.$$

Since u is entire harmonic, its expansion of the form (2.1) centered at the origin has infinite radius of convergence. Hence

$$u(z) = \sum_{k=0}^{\infty} c_k z^k + \sum_{k=1}^{\infty} c_{-k} \bar{z}^k, \quad z \in \mathbb{C},$$

where

$$c_k = \frac{\partial^k u(0)}{k!}, \quad c_{-k} = \frac{\bar{\partial}^k u(0)}{k!}, \quad k \geq 1.$$

Define the auxiliary functions

$$\phi(z) = r \cosh z + i \frac{1}{r} \sinh z, \quad z \in \mathbb{C},$$

and

$$F(z) = \sum_{k=0}^{\infty} c_k \phi(z)^k + \sum_{k=1}^{\infty} c_{-k} \overline{\phi(\bar{z})}^k, \quad z \in \mathbb{C}.$$

Since ϕ is entire analytic and the above series has infinite radius of convergence, the function F is also entire analytic. By assumption, $u(\phi(x)) = 0$ for all $x \in \mathbb{R}$, so it follows that

$$u(\phi(x)) = \sum_{k=0}^{\infty} c_k \phi(x)^k + \sum_{k=1}^{\infty} c_{-k} \overline{\phi(x)}^k = F(x) = 0.$$

Therefore $F(x) = 0$ for all $x \in \mathbb{R}$. By the uniqueness of analytic continuation, we have $F(z) = 0$ for all $z \in \mathbb{C}$.

By periodicity properties $\cosh(z + i\pi) = -\cosh z$ and $\sinh(z + i\pi) = -\sinh z$, we have that

$$\phi(z + i\pi) = -\phi(z) = \phi(z - i\pi), \quad z \in \mathbb{C}.$$

Let $x \in \mathbb{R}$, notice that

$$\phi(x + i\pi) = -r \cosh x - i \frac{1}{r} \sinh x \in H_r^-.$$

Therefore the expansion

$$\begin{aligned} u(\phi(x + i\pi)) &= \sum_{k=0}^{\infty} c_k \phi(x + i\pi)^k + \sum_{k=1}^{\infty} c_{-k} \overline{\phi(x + i\pi)}^k \\ &= \sum_{k=0}^{\infty} c_k \phi(x + i\pi)^k + \sum_{k=1}^{\infty} c_{-k} \overline{\phi(x - i\pi)}^k \\ &= \sum_{k=0}^{\infty} c_k \phi(x + i\pi)^k + \sum_{k=1}^{\infty} c_{-k} \overline{\phi(x + i\pi)}^k \\ &= F(x + i\pi) = 0. \end{aligned}$$

Hence $u(z) = 0$ for all $z \in H_r^-$. We have proved that u vanishes on both connected components H_r^+ and H_r^- of $\mathcal{H}_r$. Therefore $u(z) = 0$ for all $z \in \mathcal{H}_r$, and hence for all $z \in \mathcal{H}$. $\square$

Remark. Theorem 10 shows that for entire harmonic functions, the vanishing condition propagates from one hyperbolic arc to the whole hyperbola. This phenomenon is only on the entire setting and does not hold for locally defined harmonic functions. On the other hand, Theorem 10 does not imply that u vanishes identically in $\mathbb{C}$. For instance, the real-valued entire harmonic function

$$u(x + iy) = xy - 1$$

vanishes precisely on the hyperbola $xy = 1$. For related symmetry-based results for other planar curves, see Khavinson and Shapiro [4, Proposition 3].

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MATHEMATICS, FACULTY OF SCIENCE, CENTRE FOR MATHEMATICAL SCIENCES, LUND UNIVERSITY, P.O. BOX 118, SE-221 00 LUND, SWEDEN
Email address: ru0361wa-s@student.lu.se